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Summary

This docs contains the reference to prepare and configure the infrastructure that is to be further onboarded into pcd for creating host clusters and catering the VM workloads.

Day-1 Operation

Preparing the baremetal hosts

Prepare the baremetal hosts by installing the OS. the Boot disk should be taken a minimum as 500GB.

At the time of preparation following resources should be in place to be utilised during PCD host preparation and rollout:

- Netapp Volume created and ready via a dedicated SVM
  - It should have a valid SSL certificate installed.
  - The details of the SVM like the IP address, vserver name, volume names etc should be handy
- Likewise details of any other storage backend that will be used for PCD storage should be known and ready
- A dedicated NFS mount point should be prepared and ready for to be used for PCD image services.
  - This mount point should be capable to be mounted and available by `pf9` user on every hosts
- underlying network VLAN that will be used for the PCD operation should be known and ready.

> ⚠ **Netapp backend storage configuration requirements**
>
> - Ensure the SVM volumes are provisioned from different aggregates for resiliency
> - The SVM volumes that are provisioned to be used as cinder backends should have symmetric volume size.

OS version

- The current PCD version is certified with **Ubuntu 24.04LTS** release as the base OS version

Pre-requistes for all the hosts to be part of PCD cluster

Host hardening requirements

platform9 services on the host works compliantly with CIS benchmark methodologies but there are few exception that needs to ensure the overall operation of pcd service are smooth and functioning:

- Ensure immutable bit is not set on the `/etc/hosts` file. **This file will be updated by the hostagent process during role apply operation.**
- although IPv6 module has no direct dependency with the PCD service, consider it enabling for satisfying any application connectivity dependencies.
- Ensure the non-required user groups like `games, uucp, lp` etc are properly commented or removed from `/etc/group` and `/etc/gshadow` files as **difference in these two group files can sometime lead to apt package installation failures.**

Host level local users

- Ensure each host that is going to part of PCD cluster ensure to have a **user** which has password-less sudo access which can be access via ssh private key.
- There is a reference script in case required in the reference section that can be used to setup password less sudo user

Interfaces/network configs

- Setup the server level pre-requistes: https://docs.platform9.com/private-cloud-director/getting-started/pre-requisites on very host that is to be onboarded.
- Prepare and apply netplan configuration for every host to be used.
- A sample netplan:

```
network:
  version: 2
```

```yaml
  ethernets:
    enp24s0f0:
      dhcp4: no
    enp24s0f1:
      dhcp4: no
    enp59s0f0:
      dhcp4: no
    enp59s0f1:
      dhcp4: no
  bonds:
    bond0:
      interfaces:
        - enp24s0f0
        - enp24s0f1
      mtu: 9000
      parameters:
        mode: 802.3ad
        down-delay: 0
        lacp-rate: fast
        mii-monitor-interval: 100
        transmit-hash-policy: layer3+4
        up-delay: 0
    bond1:
      interfaces:
        - enp59s0f0
        - enp59s0f1
      mtu: 9000
      parameters:
        mode: 802.3ad
        down-delay: 0
        lacp-rate: fast
        mii-monitor-interval: 100
        transmit-hash-policy: layer3+4
        up-delay: 0
  vlans:
    vlan.301:
      id: 301
      link: bond0
      addresses:
        - xx.xx.xx.xx/18
      routes:
        - to: default
          via: xx.xx.xx.xx
      nameservers:
        addresses:
          - xx.xx.xx.xx
          - xx.xx.xx.xx
    vlan.305:
      id: 305
      link: bond1
      dhcp4: true
      dhcp6: false
      dhcp4-overrides:
        use-routes: false
        use-dns: no
    vlan.306:
      id: 306
      link: bond1
      dhcp4: false
      dhcp6: false
```

```
      dhcp4-overrides:
        use-routes: false
        use-dns: no
      addresses:
       - xx.xx.xx.xx/18
```

- ensure the DNS resolution is functional and working: `nslookup google.com` or to any fqdn
- Setup and enable NTP or `systemd.timesyncd` service and it is synced. To check, get status of the following

```
# ntpq -p
     remote           refid      st t when poll reach   delay   offset  jitter
==============================================================================


ntpdate -u 0.ubuntu.pool.ntp.org
```

Additional package installation

- Ensure following packages are installed:

```
apt install lsscsi sg3-utils multipath-tools scsitools open-iscsi multipath-tools-boot nfs-common
```

NOTE: Ensure to install on every host that will be used for PCD environment.

Storage configuration

NOTE: Ensure to install on every host that will be used for PCD environment.

Important Note

- Ensure the required packages are installed on the hosts
- The iscsi initiator name needs to be unique for every node.
- Every node need to have the iscsi login to the target portal in question.

Connecting to ISCSI store backends

- Start and enable `iscsid`:

```
sudo systemctl enable --now iscsid
sudo systemctl enable --now multipathd
```

- Set Uni IQN for every host:

```
cat /etc/iscsi/initiatorname.isci
```

- set `node.startup = automatic` in `/etc/iscsi/iscsid.conf`
- Configure `multipath.conf` file:
  - Get the `wwn` id of local disk: `/lib/udev/scsi_id -gud /dev/sda`

```
defaults {
user_friendly_names no
find_multipaths yes

}

blacklist {
```

```
wwid <wwid-of-local-disk>

}
```

- Restart `iscsid` and `multipathd` service:

```
systemctl restart iscsid
systemctl restart multipath-tools
```

- Setup the LVM filter to exclude the mpath device during boot:

```
filter = [ "a|^/dev/nvme[0-9]+n[0-9]+$|", "a|^/dev/md[0-9]+$|", "a|^/dev/md[0-9]+p[0-9]+$|", "a|^/dev/sda+[0-9]*$|", "r|.*|" ]
global_filter = [ "a|^/dev/nvme[0-9]+n[0-9]+$|", "a|^/dev/md[0-9]+$|", "a|^/dev/md[0-9]+p[0-9]+$|", "a|^/dev/sda[0-9]*$|", "r|.*|" ]
```

This Avoids ISCSI/mpath LVM scans during reboots and will pick up only the first local device.. NOTE: This filter is extremely important to be set properly unless it can lead system unable to boot.

- Update `initramFS` and reboot:

```
update-initramfs -u -k all
```

This ensure the filter is also updated inside the initramfs to make it effective

- Run the iscsiadm discovery command to fetch the current portal details:

```
iscsiadm -m discovery  -t st -p <portal-IP1>
iscsiadm -m discovery  -t st -p <portal-IP2>
```

- Once discovery is successful, login and connect to the portal :

```
iscsiadm -m node -T <target-iqn> --login
iscsiadm -m node -p <netapp-portal-ip>:3260 --op=update -n node.startup -v automatic
```

NOTE: perform this for all the targets

- Reload `multipath` service:

```
multipath -r
```

- Check `multipath` command to see if the LUNs are now visible:

```
multipath -ll
```

NOTE: With Storage backend like netapp AFF where the SVM will expose the storage as volume, the multipath command will only display the LUNs that are created and consumed by the VM

Adding FSTAB entries

NOTE: Ensure its applied for all hosts

This mount point will be serving as the backend store for the glance/image library service in the PCD framework.

- Add the NFS mount point entry in the `/etc/fstab` file to have `/var/opt/imagelibrary/data` for every host:

```
cat /etc/fstab
NFS-IP:mountpoint /var/opt/imagelibrary/data <NFS OPts>  0 0
```

NOTE: The mount point added in the `fstab` should be the one that is mentioned in the blueprint under image library section

- Create the directory `/var/opt/imagelibrary/data` path on the host and has file/folder permission to `0755`

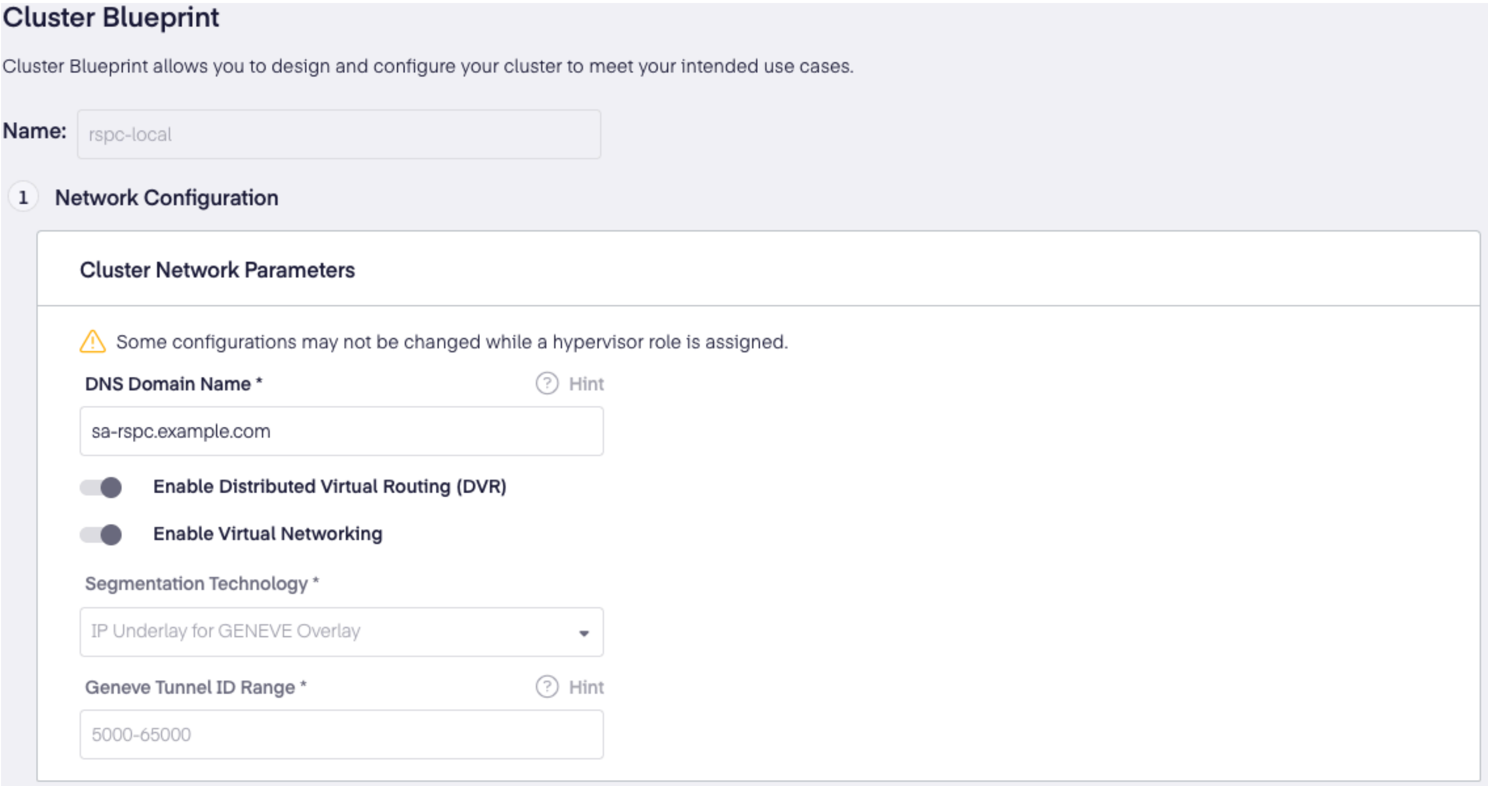
Day-2 operation

Pre-requiste

- Infrastructure storage backend ready with the details required for cinder configuration.
- All steps to prepare the hosts are completed successfully from the Day-1 requirements.
- PCD region portal is deployed and accessible via the shared credential through the platform9 personnel.
- The prefix/suffix names, abbreviation that are needed to create PCD resources like tenant, cinder backend, users etc are known and ready.
- SSO saml authentication config details to be collected and ready.

PCD UI Portal configurations

Cluster Blueprint

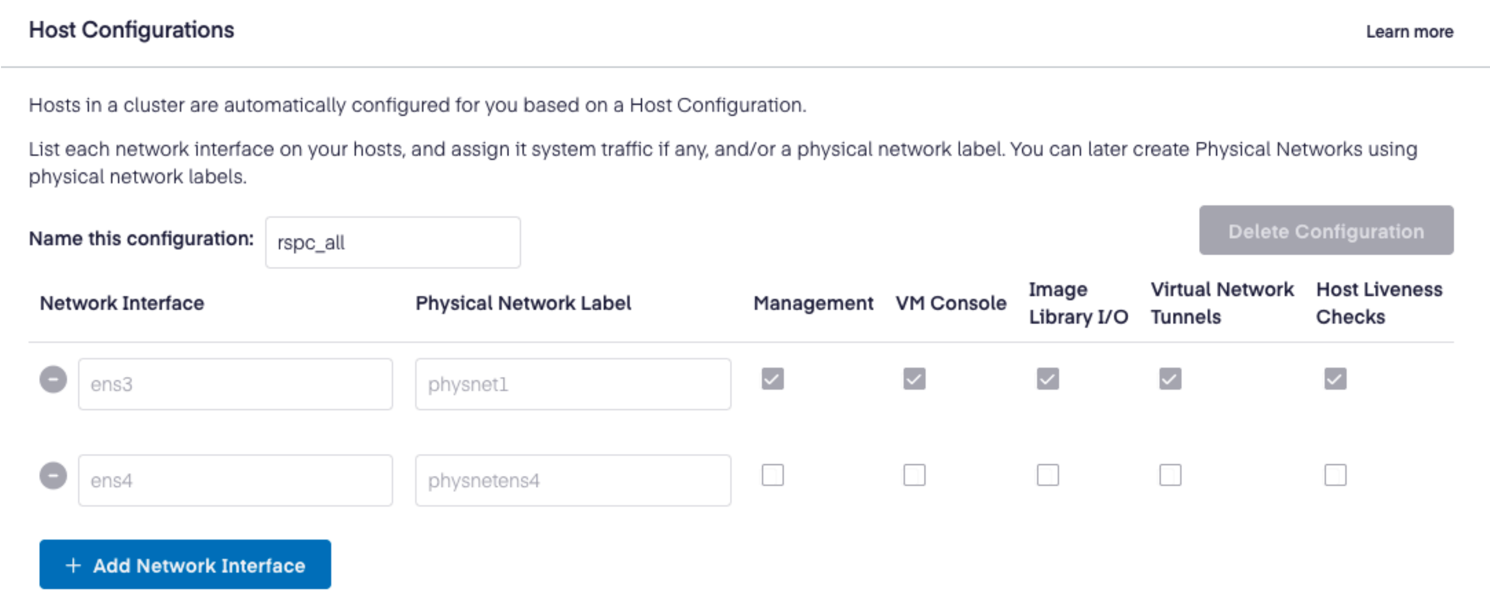


- Define the cluster name, and domain name as needed.
- DVR and virtual networking are to be left as enabled
- The Segmentation to be set as IP underlay for GENEVE overlay which will be sed for VM overlay communication.
- The GENEVE tunnel range can be set as shared

NOTE:

- The GENEVE overlay will be used for VM overlay network for connecting them to the datacenter fabric.
- This tunnel will only be used mainly during east west communication between VMs residing on different hypervisor host and should not have any impact on any existing datacenter level VXLAN tunnel etc.

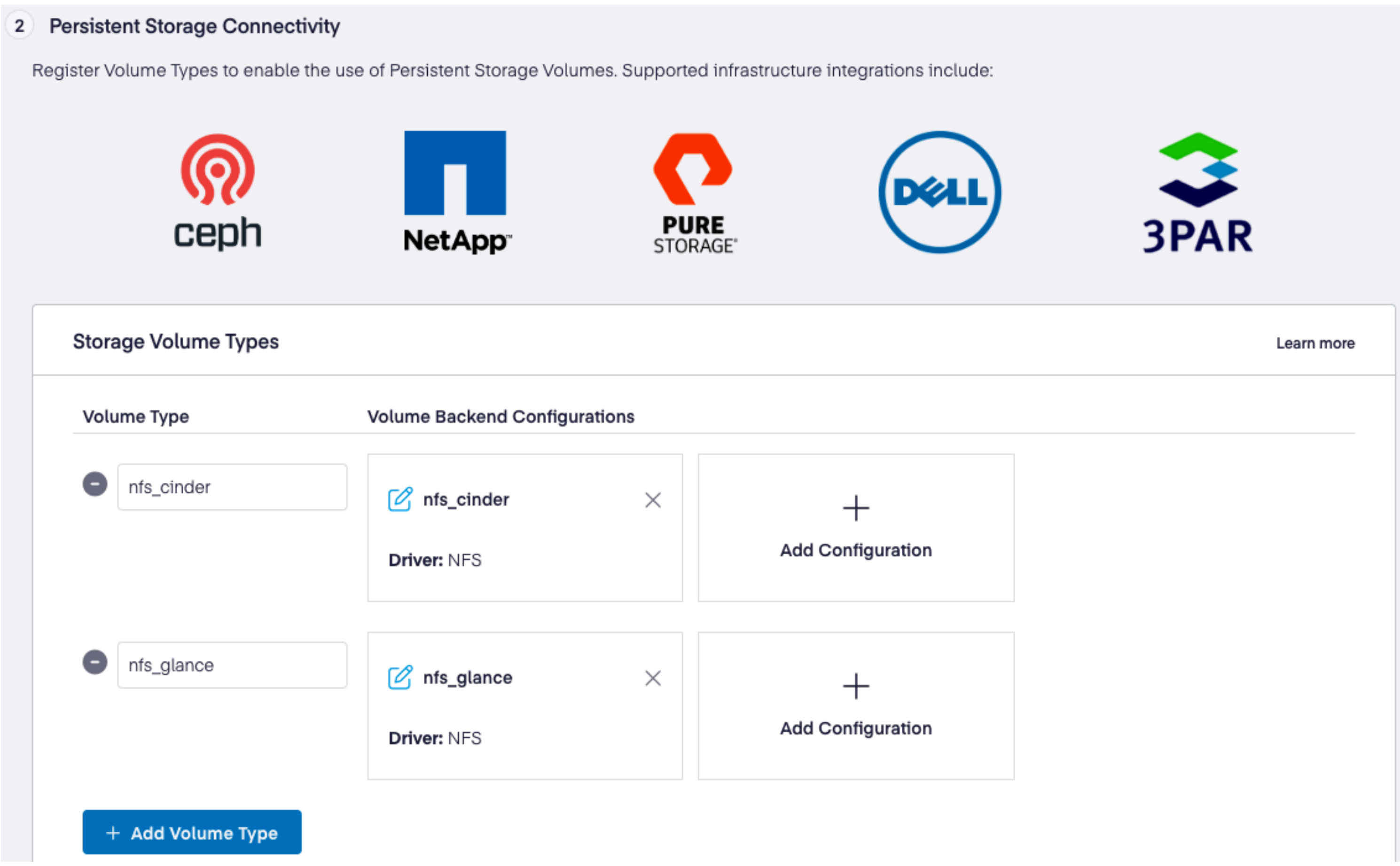
Network profile configuration



This defines the interface layout that are set on the hosts which will be part of the PCD cluster

- The interfaces defined in this within the network profile will be used by PCD for communication (service and VM connections alike).
- For interface to be considered and mapped within PCD, it would **either need to have a label set or it should have one of the PCD traffic mapped on to it.**
- Interfaces not part of this configuration section, **can continue to exists** and be used for other communications by the hosts while being part of the PCD cluster.

Storage configuration



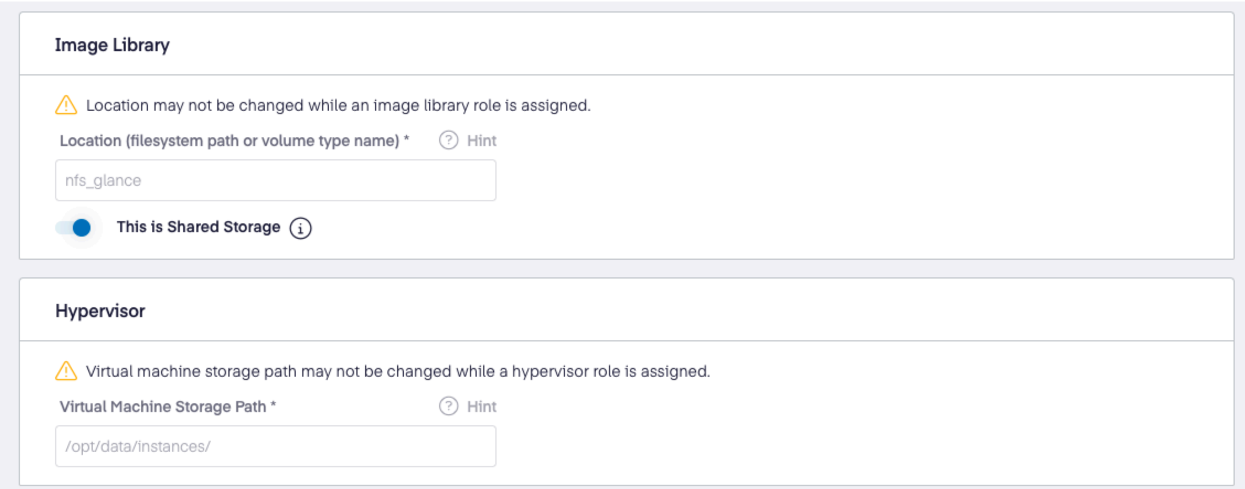
This section is used to define the backend store for the pcd block storage service.

- The `volume_type`, as the name suggest, defines the type of the storage.
- The `Volume backend section` is to be used to define the actual storage driver configuration

NOTE:

- When multiple volume backend sections are defined for a single volume Type, PCD will treat that volume_type as multi-backend operation and will use the internal algorithm to balance and place the block store requests across the them.

Image and ephemeral storage configuration



The image libary section will default to `/var/opt/imagelibrary/data` directory location which will be local on every pcd hosts that are applied with glance role.

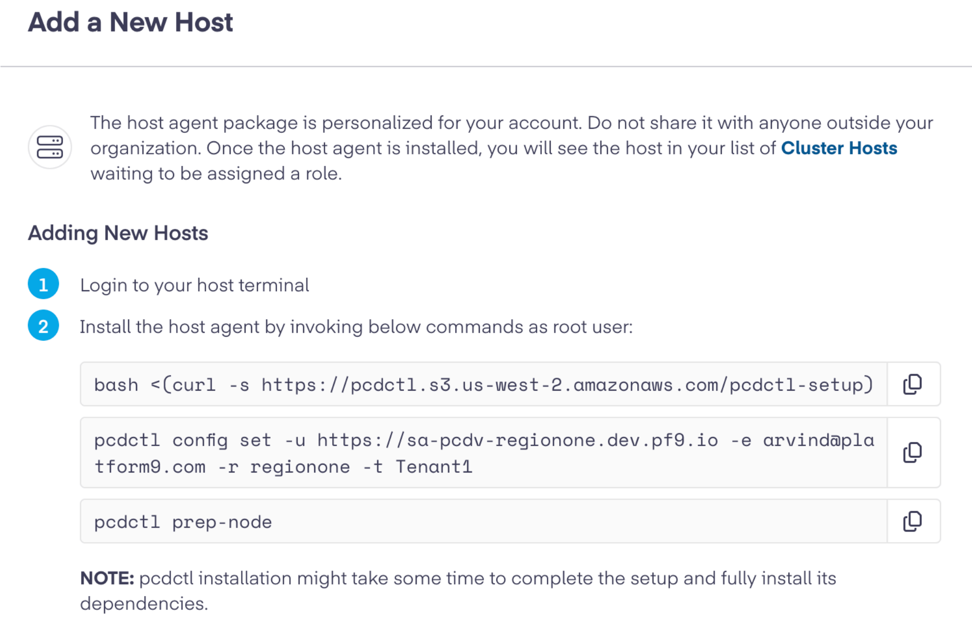
- For external nfs mount, this directory path should be noted properly and can be used to mount over external nfs mount through per-host `fstab` entry
- If the volume_type name is mentioned in this section, the PCD will configure the cinder store as the backend for image service.

Enabling the `this is shared storage` option will allow the glance role to be removed properly in situation where its applied on multiple hosts.

Host onboarding

The onboarding of prepared hosts requires following steps:

- From the PCD UI under the `cluster hosts` section Click the add hosts button
- Apply the commands per host that are mentioned in the steps page:



After successful process, the hosts would be visible in `unauthorized` state in the cluster hosts section.

Once the hosts are visible in the UI change ownership of `/var/opt/imagelibrary/data` to `pf9:pf9group`:

```
chown -R pf9:pf9group /var/opt/imagelibrary/data
```

Netapp AFF 800 specific configuration

After the host is enrolled and available in the PCD portal:

- Ensure the SVM or the netapp `vserver` name/FQDN is resolvable. If not as a workaround add an entry in the `/etc/hosts` file for all the hosts in the cluster which will be actually going to use it:

```
172.18.151.222 SVM-hostname
```

NOTE: Ensure its applied for all hosts.

- Create a directory `/opt/pf9/certs/`
- Download and save the SSL cert for storage like Netapp to the `certs` directory:

```
# openssl s_client -showcerts -connect SVM_IP_FQDN:443 </dev/null 2>/dev/null | openssl x509 -outform PEM >
/opt/pf9/certs/SVM_IP_cinder.pem
```

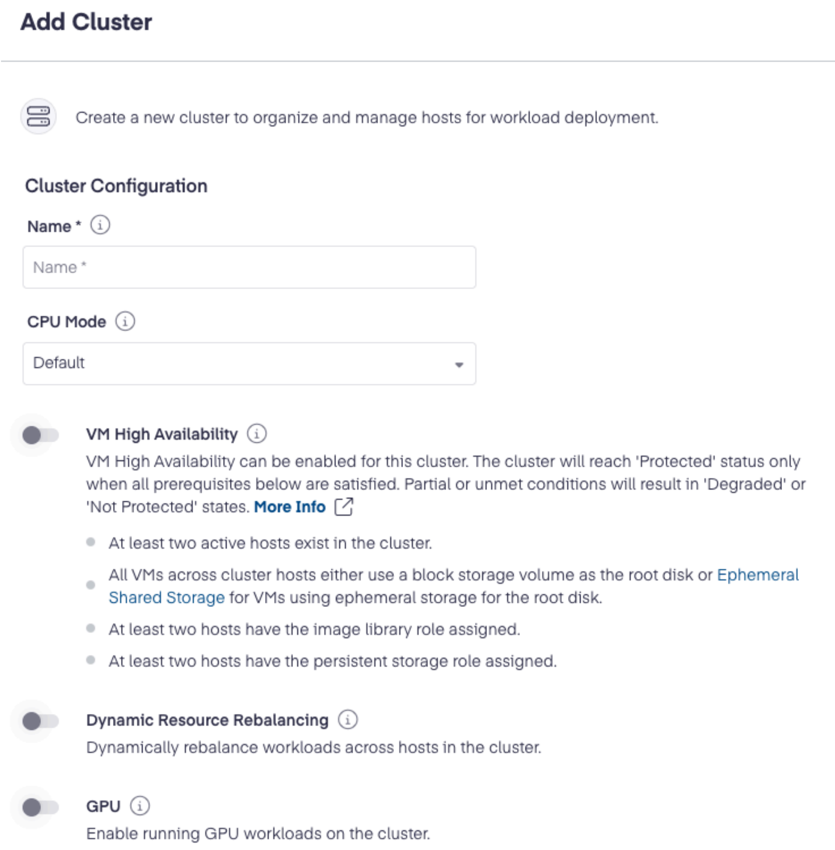
NOTE:

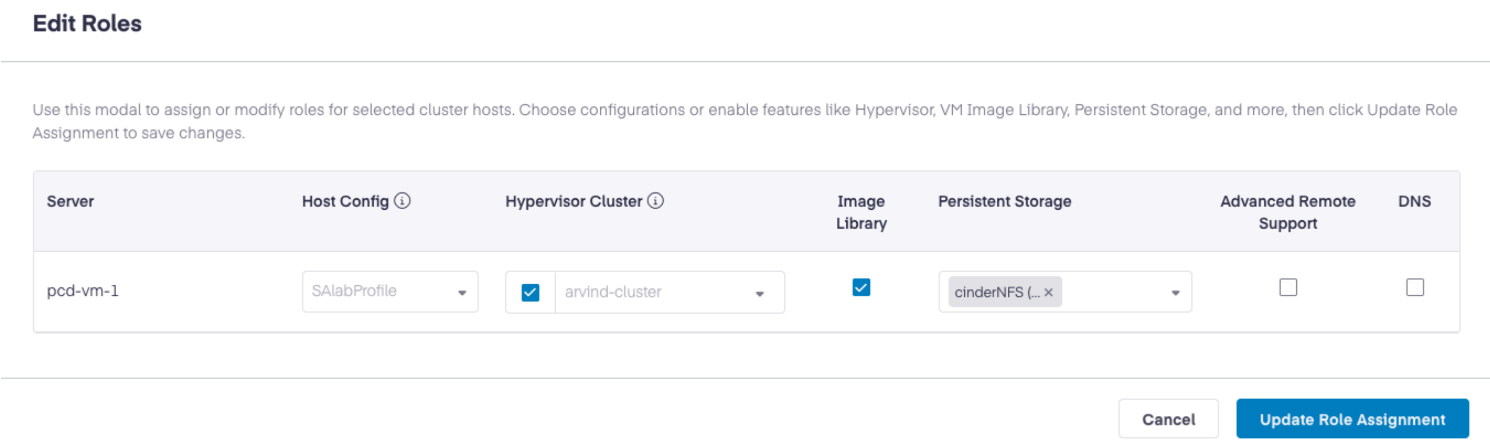
- The certificate that is created in the netapp store should have the CN name matching to the FQDN.
- Ensure its applied for hosts with `persistent-storage` role applied
- Set the UID and GID for the copied certificate to `pf9:pf9group`

```
chown pf9:pf9group /opt/pf9/etc/pf9-cindervolume-base/conf.d/SVM_IP_cinder.pem
```

Role assignment operation

Create a cluster with desired name from the clusters -> Create cluster section:





Select the host or hosts from the cluster section and click the edit role button which will have the above view

- Select the host config profile for selecting and setting the network profile defined in the blueprint section
- The Select the cluster name from the drop down in the hypervisor cluster section.
- Set the image library flag to apply the glance/image role to the host
- From the persistent storage drop down, select the required backend as per the configuration available.

Post hypervisor role apply specific configuration

This is required to ensure the `iscsi` backend is referred and used by nova over `multipath`.

- Ensure the set following in `/opt/pf9/etc/nova/conf.d/nova_override.conf` file has following volume backend configuration in place:

```
volume_use_multipath = True
```

NOTE: Need to be applied for all the hosts in the cluster that has hypervisor role.

- Restart the `pf9-ostackhost` nova service on every host in the cluster after this change:

```
systemctl restart pf9-ostackhost
```

Post Glance role apply specific configuration

- Glance backend is to be mounted via external NFS mount.
- Ensure the role is applied from the pcd portal.
- set the `/etc/fstab` entry on the **glance role applied hosts** to point to the nfs mount:

```
<IP>:MOUNT_POINT_DIR  /var/opt/imagelibrary/data nfs vers=3,proto=tcp,lookupcache=pos 0 0
```

- Mount the directory:

```
mount -a
```

- Restart the glance service on the host:

```
systemctl restart pf9-glance-api.service
```

NOTE:

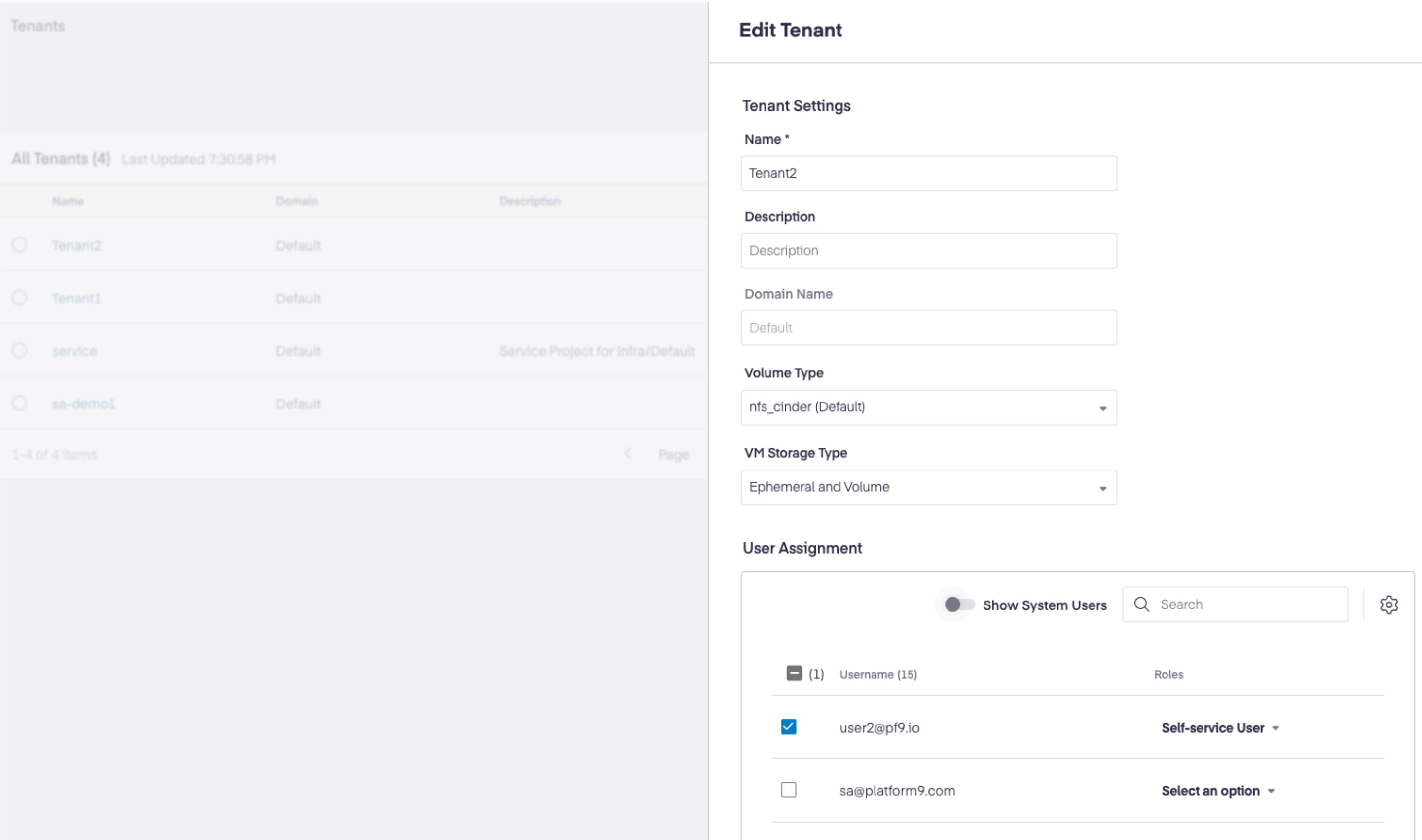
- Ensure the NFS mount is shared with proper policy to allow pf9 user to be able to write on the mount.

- Ensure the `/var/opt/imagelibrary/data` has local directory has the proper permission (`0755`) set to allow the mount to be visible and accessible by `pf9` user.

Setting up resources post host cluster creation

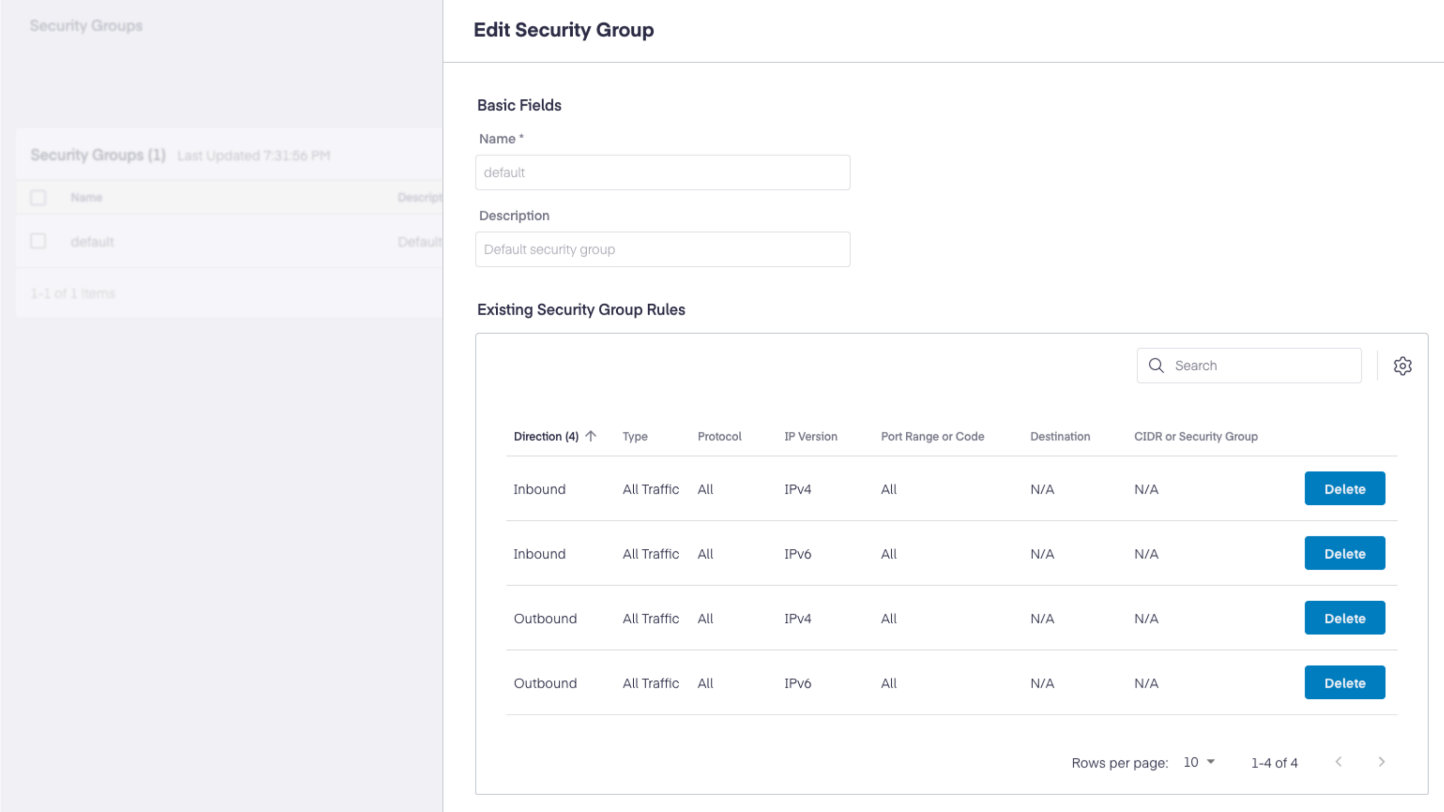
Tenant creation

- From the UI select the **users and tenants** section to create the tenant as needed and ensure to map the required users with the needed RBAC roles (self-service/admin/read-only).

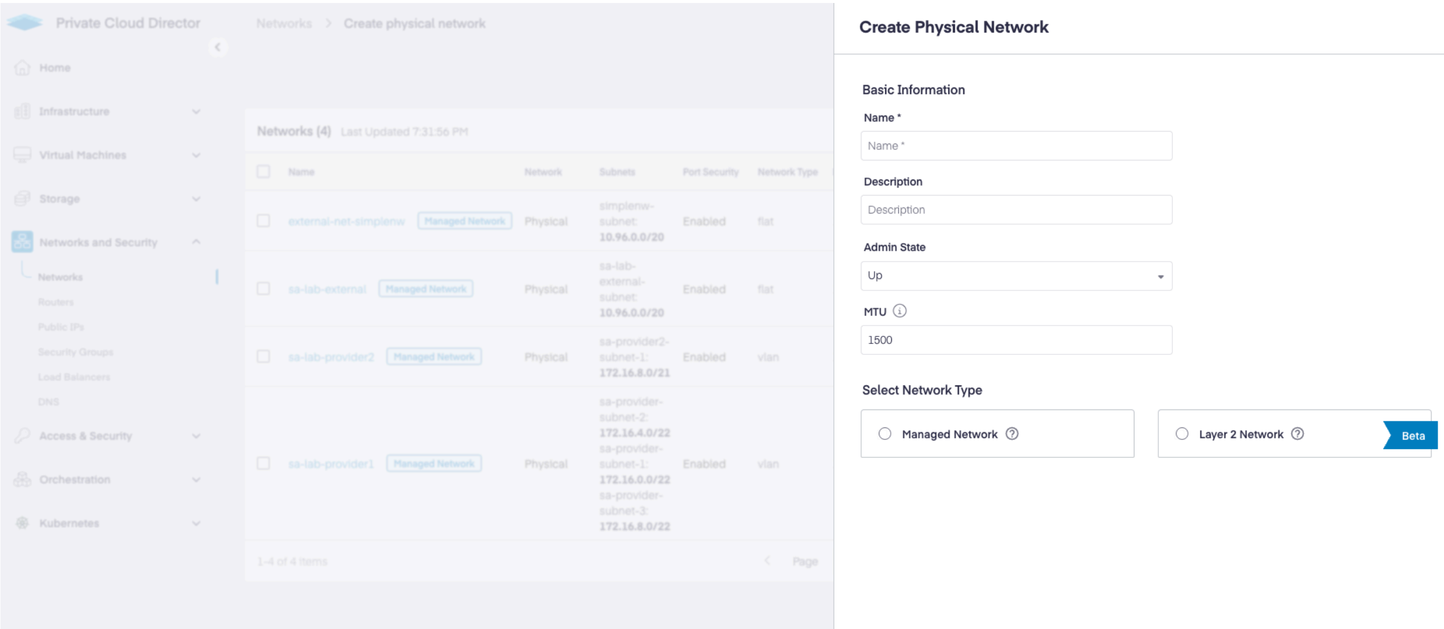


Provider network creation

- Select the Security group and edit the default security group to allow all traffic for inbound and outbound options:



- Under the **Networks and Security** tab select option `create network -> physical network` to setup FLAT or VLAN based networks:



NOTE:

- Ensure the port security is enabled for the network/ports with the default security group applied. This is highly recommended as it provided options like reverse path filtering, block improper mac learning scenario etc.
- DHCP on the neutron is recommended to be enabled to ensure proper working of services like cloud-init needed for VM customisation. To avoid any conflicts with other running dhcp solutions in a provider network, the IP allocation pool for the neutron subnet can be set with minimum 2 free IPs from the pool. The first IP from this pool will be used for the DHCP server within neutron.

Image upload operation

From the UI the images up to size 5GB can be uploaded. For any large size images, the openstack cli should be used. To perform such upload operation:

- Download the `openstack.rc` file from the UI from the **api access -> pcdctl rc section** and save it as `somename.rc` file on a system where openstack cli is installed and ready.

- Source the rc file:

```
source somefile.rc
```

- Upload the image using the command:

```
openstack image create --os-interface admin  --insecure --container-format bare --disk-format qcow2  --private --file somefile.img <upload-filename>
```

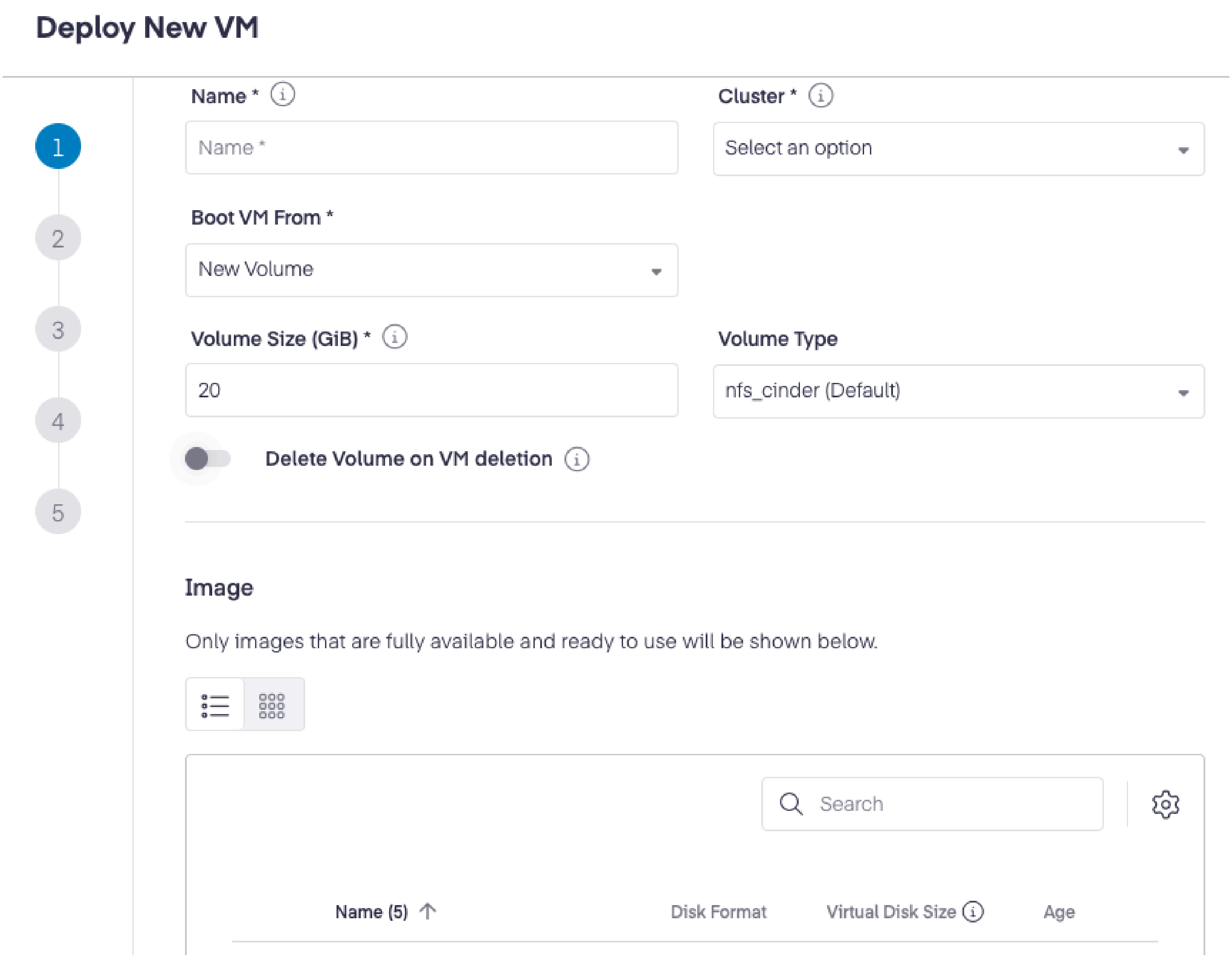
In use cases like launch VJB VMs from images, it is recommended to apply below properties to the vjb image during upload operation:

```
openstack image create --os-interface admin  --insecure --container-format bare --disk-format qcow2  --public --file somdisk.img --property hw_disk_bus=scsi   --property hw_scsi_model=virtio-scsi  --property hw_machine_type=q35 --property os_type=linux some_file_name
```

VM deployment

Pre-requiste

- Tenant is created and ready
- provider/virtual network created and ready
- Required image is uploaded and ready
- PCD host cluster is ready and functional
- Storage backend is configure and functional
- From the UI the VM can be deployed by the user:



References

Scripted way of Creating password less sudo user for host

- From any remote ubuntu host generate a ssh private public key from the command line terminal with : `ssh-keygen` command (press **enter** for all the question prompted)
- under the `.ssh/` directory of the user where the keygen command was executed, note down/copy the content of `id_ed25519.pub` file
- copy the below script on the **host** to be added in pcd locally:

```
#!/usr/bin/env bash
# Usage: sudo ./create_pwless_sudo_user.sh <username>
# EMBEDDED PUBLIC KEY - place your public key in this variable

set -euo pipefail

if [[ $# -ne 1 ]]; then
    echo "Usage: $0 <username>"
    exit 1
fi

USER_NAME="$1"

# Replace with YOUR actual public key (ssh-ed25519/id_rsa.pub content)
EMBEDDED_PUBKEY="ssh-ed25519 somekeys ubuntu@ubuntu-host"

# Must be run as root
if [[ "$(id -u)" -ne 0 ]]; then
    echo "Run as root (sudo)." >&2
    exit 1
fi

# Validate Ubuntu sudo group
if ! getent group sudo >/dev/null; then
    echo "Ubuntu 'sudo' group required." >&2
    exit 1
fi

# User handling
if ! id "$USER_NAME" &>/dev/null; then
    useradd -m -s /bin/bash "$USER_NAME"
    echo "Created '$USER_NAME'."
else
    echo "Using existing '$USER_NAME'."
fi

# Sudo group (idempotent)
if ! groups "$USER_NAME" | grep -qw sudo; then
    usermod -aG sudo "$USER_NAME"
    echo "Added to sudo group."
fi

# Passwordless sudo
SUDOERS_FILE="/etc/sudoers.d/${USER_NAME}"
echo "${USER_NAME} ALL=(ALL) NOPASSWD: ALL" > "$SUDOERS_FILE"
chmod 440 "$SUDOERS_FILE"
if command -v visudo >/dev/null && ! visudo -cf "$SUDOERS_FILE"; then
    echo "Sudoers error." >&2
```

```
    rm "$SUDOERS_FILE"
    exit 1
fi
echo "Passwordless sudo set"

# SSH KEY SETUP with EMBEDDED KEY
SSH_DIR="/home/${USER_NAME}/.ssh"
AUTH_KEYS="${SSH_DIR}/authorized_keys"

mkdir -p "$SSH_DIR"
chown "${USER_NAME}:${USER_NAME}" "$SSH_DIR"
chmod 700 "$SSH_DIR"

# Use embedded key (always)
if [[ -z "$EMBEDDED_PUBKEY" || "$EMBEDDED_PUBKEY" == "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAI... your-
email@example.com" ]]; then
    echo "UPDATE REQUIRED: Replace EMBEDDED_PUBKEY with your actual public key!" >&2
    exit 1
fi

# Append embedded key if not present
if ! grep -F "$EMBEDDED_PUBKEY" "$AUTH_KEYS" >/dev/null 2>&1; then
    echo "$EMBEDDED_PUBKEY" >> "$AUTH_KEYS"
    echo "SSH key added."
else
    echo "key already exists."
fi

chown "${USER_NAME}:${USER_NAME}" "$AUTH_KEYS"
chmod 600 "$AUTH_KEYS"

echo
echo "'$USER_NAME' READY WITH EMBEDDED ACCESS!"
echo "- Passwordless sudo set"
echo "- SSH key added successfully"
```

- Copy the public key collected earlier and add it in the script in the `EMBEDDED_PUBKEY="ssh-ed25519 somekeys ubuntu@ubuntu-host"` **(replace the values here for the variable)**
- Set the execute permission to the script: `chmod a+x password-less_usercreate.sh` and execute it `./password-less_usercreate.sh someuser

This will create the user if not present with password less option and it can be used to ssh from remote using the private key.

NOTE:

- Ensure the **private key** on the remote ubuntu host that was generated, is kept safe and handy which is needed to access over ssh.

Storage role specific config reference

- To setup cinder backend the blueprint in the region portal, set the preferred volume backend name and in the configuration of the backend select the option `custom` driver and set the driver as: `cinder.volume.drivers.netapp.common.NetAppDriver`. Following is the actual sample of `cinder.conf` file that will be applied in every host that has `persistent-storage` role applied from the UI:

```
[SOME-VOLUME-TYP-NAME]
volume_driver = cinder.volume.drivers.netapp.common.NetAppDriver
volume_backend_name = Some-backend-name
pf9_identifier_timestamp = 1769089443.246261
```

```
volumes_dir = /opt/pf9/etc/pf9-cindervolume-base/volumes/
netapp_login = vsadmin
netapp_vserver = SVM-name
netapp_password = secret://8b128daa-1a83-45f9-b157-2758831c0212
netapp_server_port = 443
volume_use_multipath = true
netapp_storage_family = ontap_cluster
netapp_transport_type = https
netapp_server_hostname = <SVM-name>
netapp_storage_protocol = iscsi
netapp_pool_name_search_pattern = SVM-name
netapp_driver_reports_provisioned_capacity = true
netapp_ssl_cert_path = /opt/pf9/certs/<SVM-name>=new.pem
netapp_certificate_host_validation = false
netapp_use_legacy_client = true
```